

A Novel Ferroelectric Tunnel FET-Based Synapse With Complementary Hebbian Plasticity Enhancing Learning of Spiking Neural Networks

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Abstract—In this work, for the first time, a novel ferroelectric (FE) tunnel FET (FeTFET) based synapse device with complementary Hebbian plasticity is proposed and experimentally demonstrated. The device leverages FE polarization switching dynamics to achieve spike-timing-dependent plasticity (STDP) and utilizes FE-modulated ambipolar band-to-band tunneling (BTBT) transports with opposite monotonicity for complementary synaptic weight updates. The fabricated FeTFET based on the 14nm Fin-FET platform enables configurable anti-STDP and STDP modes with single transistor of high CMOS compatibility, facilitating advanced supervised and reinforcement learning. Moreover, benefiting from the transition of BTBT windows, the FeTFET synapse exhibits historical activity-aware plasticity, enhancing the stability of the learning process and demonstrating its potential for highly integrated neuromorphic computing systems.

Index Terms—Ferroelectric tunnel FET (FeTFET), complementary synaptic plasticity, spiking neural networks.

I. INTRODUCTION

BRAIN inspired spiking neural networks (SNNs), regarded as the third generation of neural networks, has attracted significant attention due to potential learning capabilities that resemble human learning processes [1] (Fig. 1a). The spike-timing-dependent plasticity (STDP) function of fundamental synaptic elements has been extensively studied and has been demonstrated to effectively implement Hebbian learning rules, enabling basic unsupervised learning in SNNs [2]. In Hebbian STDP (Fig. 1b), when pre-synaptic pulses occur before

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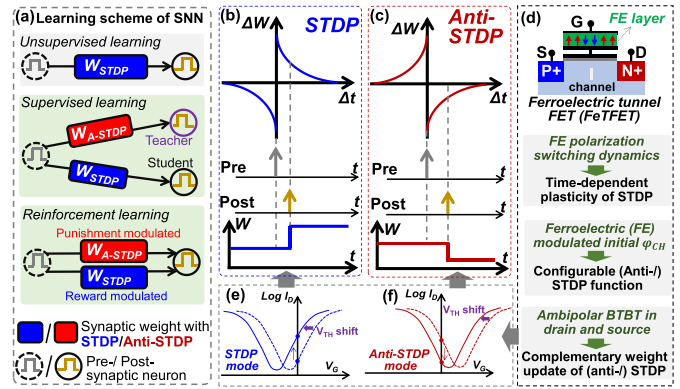


Fig. 1. (a) The learning schemes of SNNs, including basic supervised learning, advanced unsupervised learning using remote supervised methods, and reinforcement learning with reward-modulated STDP algorithms. The synaptic weight behavior of (b) STDP and (c) anti-STDP behaviors are both necessary for advanced learning, which realize complementary Hebbian plasticity. (d) The structure of the FeTFET device with ambipolar BTBT transport mechanism, and its principles of complementary weight update for (e) STDP and (f) anti-STDP mode.

post-synaptic pulses, synaptic weight increases, corresponding to long-term potentiation (LTP) process. Conversely, when pre-synaptic pulses follow post-synaptic pulses, long-term depression (LTD) occurs [3]. This behavior has been emulated by various non-volatile devices such as memristors [4], ferroelectric FETs (FeFET) [5], among others. However, conventional Hebbian STDP alone is insufficient for complex learning tasks in SNNs. Neuroscience research indicates that anti-Hebbian STDP behavior (Fig. 1c), which is the complementary function of Hebbian STDP and features synaptic plasticity in the opposite direction under the same stimulus, is essential for human learning processes involving interaction with the external environment [2]. This complementary Hebbian plasticity can enable advanced supervise learning [6], [7] and reinforcement learning [8], [9], [10], [11] (Fig. 1a). Two-terminal memristors have been used to implement complementary STDP, requiring separated write and readout path along with additional complementary cells and periphery

circuit for polarity regulation of the synaptic modulation pulses [6], [9]. Recently, three-terminal non-volatile devices with decoupled write-readout path have been explored for anti-STDP implement [7], [10], [11], [12]. These devices with back-gate structure rely on the bipolar transport properties of two-dimensional materials, such as graphene [7], WSe_2 [11], carbon nanotubes [10], to achieve complementary weight updates, still facing challenges regarding high synaptic operation voltage, reliability issues, CMOS compatibility and large-scale integration. In our previous work, we demonstrated that the CMOS compatible ferroelectric tunnel FET (FeTFET) exhibits translatable ambipolar transport properties [13].

In this work, a novel ferroelectric tunnel FET (FeTFET) based configurable synapse device is proposed and experimentally demonstrated. It enables interchangeable dynamic modes of STDP and anti-STDP for complementary Hebbian learning within a single FeTFET with high COMS compatibility (Fig. 1d-f). Additionally, the FeTFET can simultaneously realize advanced activity-aware plasticity for stabilized learning, showing great potential for highly-integrated SNN systems.

II. RESULTS AND DISCUSSIONS

A. Experiments of FeTFET for Complementary Synapse

The FeTFET structure has a ferroelectric (FE) layer inserted into gate stack of conventional tunnel FET (TFET) which is a gate-controlled reversible biased P-I-N structure (Fig. 1d). In this work, the TFET is experimentally fabricated on the foundry platform of 14 nm FinFET technology node, which inherently offers well-controlled process stability and variation. It exhibits distinct ambipolar band-to-band tunneling (BTBT) current at both source and drain junctions (Fig. 2a), owing to the device layout design without drain underlap region. The experimental FeTFET is constructed by connecting a FE capacitor with 10nm $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ FE layer (Fig. 2b) to the gate of TFET. The FE capacitor with relatively small area is selected to achieve capacitance matching, ensuring a higher voltage drop across the FE layer. When a sweep voltage of large amplitude range is applied to the gate of FeTFET (Fig. 2c), the I_D - V_G curves under quasi-DC measurement exhibit a typical ferroelectric hysteresis behavior with ambipolar characteristics, due to both the gate-voltage-modulated channel surface potential (φ_{CH}) and the polarization switching dynamics. In the experimental FeTFET with an MFIS architecture, the hysteresis window is attributed not only to ferroelectric polarization switching but also to ferroelectric-assisted charge trapping from the gate side [14], [15]. Moreover, by applying write voltage (V_{write}) pulses at the gate of FeTFET (Fig. 2d), the φ_{CH} I_D are modulated and retained, indicating the long-term plasticity of synapse. As shown in Fig. 2e, the small-amplitude read voltage has minimal effect on modulating φ_{CH} since it primarily drops across the FE layer. This behavior differs from the FeTFET design in our previous work [13], where it was optimized for translating ambipolar I_D - V_G curves.

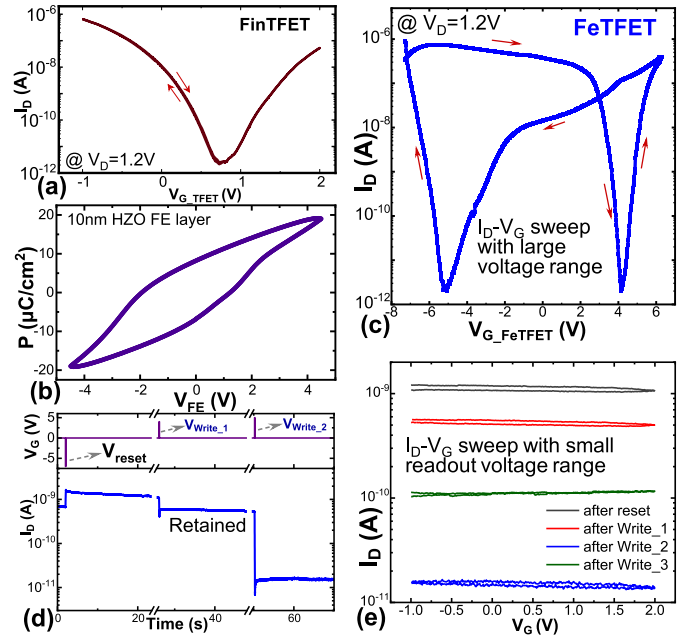


Fig. 2. (a) The I_D - V_G characteristics of the fabricated FinFET device, showing typical ambipolar behavior (b) The P-V loop of the fabricated FE capacitor with a TiN/ $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ /Pt stack structure. (c) The I_D - V_G curves of FeTFET, constructed by connecting the FE capacitor to the gate of FinFET. It is a quasi-static DC voltage sweep. The size of FE capacitor is $4\mu\text{m}^2$. The channel length of FinFET is designed to be about 30nm through the layout design. The number of fin is 10. (d) The plasticity and retention characteristics of FeTFET for synaptic weight modulation. (e) The readout I_D - V_G curves of FeTFET under small voltage.

B. Complementary STDP Behavior of FeTFET Synapse

The synaptic weight modulation scheme in the learning processes of SNNs is shown in Fig. 3a [4], [5]. The width and polarity of pulse stimuli for synaptic update are determined by the time interval (Δt) between the pre- and post-synaptic pulses. The LTP and LTD are induced by applying pulse stimuli at the gate of FeTFET synapse device.

For anti-STDP mode, the FeTFET is initially configured into a negative polarization state, where the majority of the FE polarization is oriented upward (Fig. 3b). In this state, the φ_{CH} is relatively high, activating the BTBT window at the drain junction (Fig. 3c). When continuous positive pulse stimuli are applied for $\Delta t > 0$ (Fig. 3d), partial FE polarization gradually switches to a downward state, causing increase in φ_{CH} and effective BTBT distance at drain junction. The synaptic current (I_D) measured at the interval of gate pulses decreases with the number of stimuli, indicating the LTD process in anti-STDP mode. Moreover, as the pulse width increases, the rate of synaptic I_D change per pulse increases, which is manifested as an increase in the slope of the LTD process (Fig. 3d). It is attributed to the time-dependent FE polarization switching dynamics [16], [17]. Consequently, as shown in the fourth quadrant of Fig. 3f, the synaptic weight update (ΔI_D) exhibits an inverse correlation with Δt . Conversely, when negative pulse stimuli are applied to the gate of FeTFET synapse for $\Delta t < 0$, the change in synaptic I_D follows a similar pattern as for $\Delta t > 0$ (Fig. 3e), contributing to the LTP process in the second quadrant of anti-STDP mode (Fig. 3f).

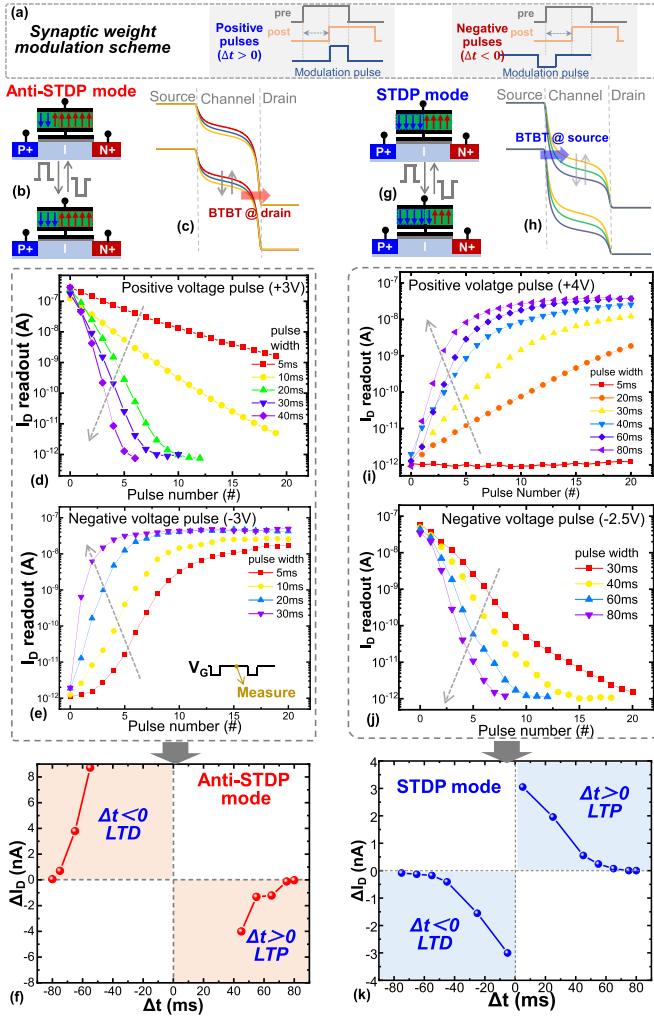


Fig. 3. Measured anti-STDP and STDP behavior of the FeTFET synapse. (a) The polarity and width of synaptic modulation pulses are determined by the timing of pre- and post-synaptic pulses. The schematic diagrams of polarization state and channel energy band for (b-c) anti-STDP and (g-h) STDP modes. The readout I_D under continuous positive/negative pulses with varying pulse widths applied to the gate of FeTFET for (d-e) anti-STDP and (i-j) STDP modes. Measured synaptic weight updates as a function of time difference at (f) anti-STDP and (k) STDP modes.

For STDP mode, the FeTFET synapse is initially configured with a primary upward polarization state, resulting in low φ_{CH} and a BTBT window at the source junction (Fig. 3gh). When applying positive (Fig. 3i) or negative (Fig. 3j) pulse stimuli in the case of $\Delta t > 0$ or $\Delta t < 0$ respectively, the direction of synaptic weight update is opposite to that of anti-STDP mode. This behavior is due to the non-monotonic ambipolar transport characteristics caused by complementary BTBT in the source and drain junction. The measured relationship between ΔI_D and Δt is depicted in the first and third quadrants in Fig. 3k, corresponding to the LTP and LTD processes. Therefore, by synergistically leveraging the time-dependent multi-domain switching dynamics of ferroelectric and complementary ambipolar BTBT mechanism of TFET, the proposed FeTFET synapse device can realize interchangeable anti-STDP and STDP behavior within a single transistor, ensuring high CMOS compatibility. It enables complementary Hebbian

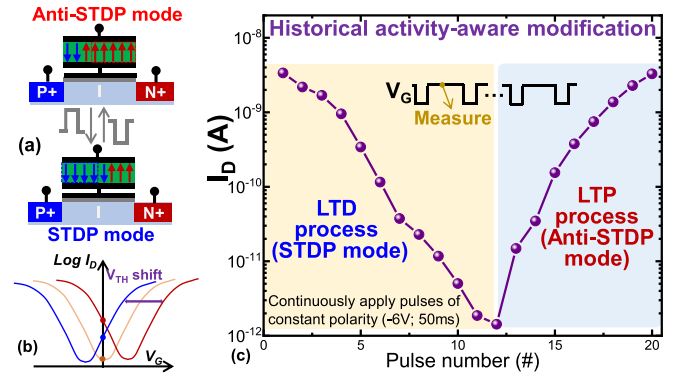


Fig. 4. (a) Schematic diagram of the polarization switching process and (b) the threshold shift when the FeTFET-based synapse transitions between anti-STDP and STDP modes. (c) Measured switching process from STDP to anti-STDP mode under continuous strong stimulus. It can emulate the historical activity-aware plasticity of biological synapse.

learning function for supervise learning and reinforcement learning of SNNs (Fig. 1a) [7], [11].

C. Historical Activity-Aware STDP of FeTFET Synapse

In addition to the complementary STDP mode, the synapse of brains also exhibits historical activity-aware plasticity [18], [19], [20], where the direction of synaptic updates also depends on the accumulated stimulus intensity received. This is specifically demonstrated by a transition from depression mode to potentiation mode when the stimulus intensity surpasses a certain threshold. This behavior can also be emulated in FeTFET synapse by utilizing the transition process between STDP and anti-STDP mode. As shown in Fig. 4, when continuous and intense synaptic modulation stimuli are applied to the FeTFET, a significant portion of the FE polarization switches direction. This causes the BTBT window to shift from the drain junction to the source junction, resulting in a transition from the LTD process to the LTP process within the synapse. This mechanism provides a stabilizing effect by preventing excessive synaptic strengthening or weakening, such as runaway activation or complete silence [20], ensuring the learning processes of SNNs. The linearity of the FeTFET synapse can be further optimized by our previous proposed hybrid gate design that broadens the coercive voltage distribution of FE layer [21].

III. CONCLUSION

This work reports a novel FeTFET based configurable synapse device for the first time, in which interchangeable STDP and anti-STDP modes are experimentally demonstrated in one FeTFET device based on 14nm FinFET technology node with high COMS compatibility. Additionally, the FeTFET can simultaneously realize advanced activity-aware plasticity for stabilized learning process, showing great potential for highly-integrated SNN systems.

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